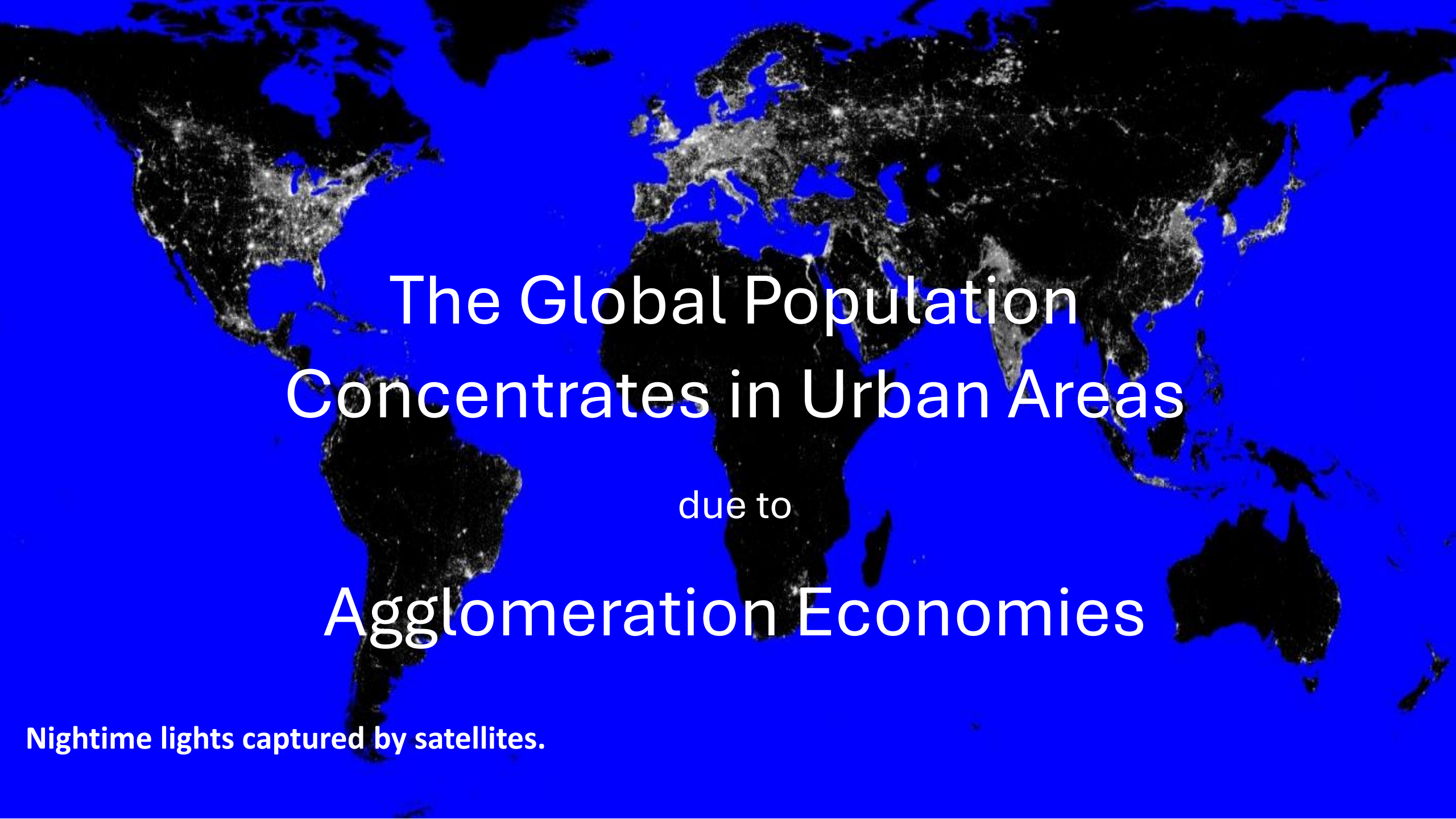
Innovation in social networks

Balázs Lengyel

Economic Complexity, Class 6 – Oct 20 2026

Social Data Science, Corvinus University of Budapest



The Global Population Concentrates in Urban Areas

due to

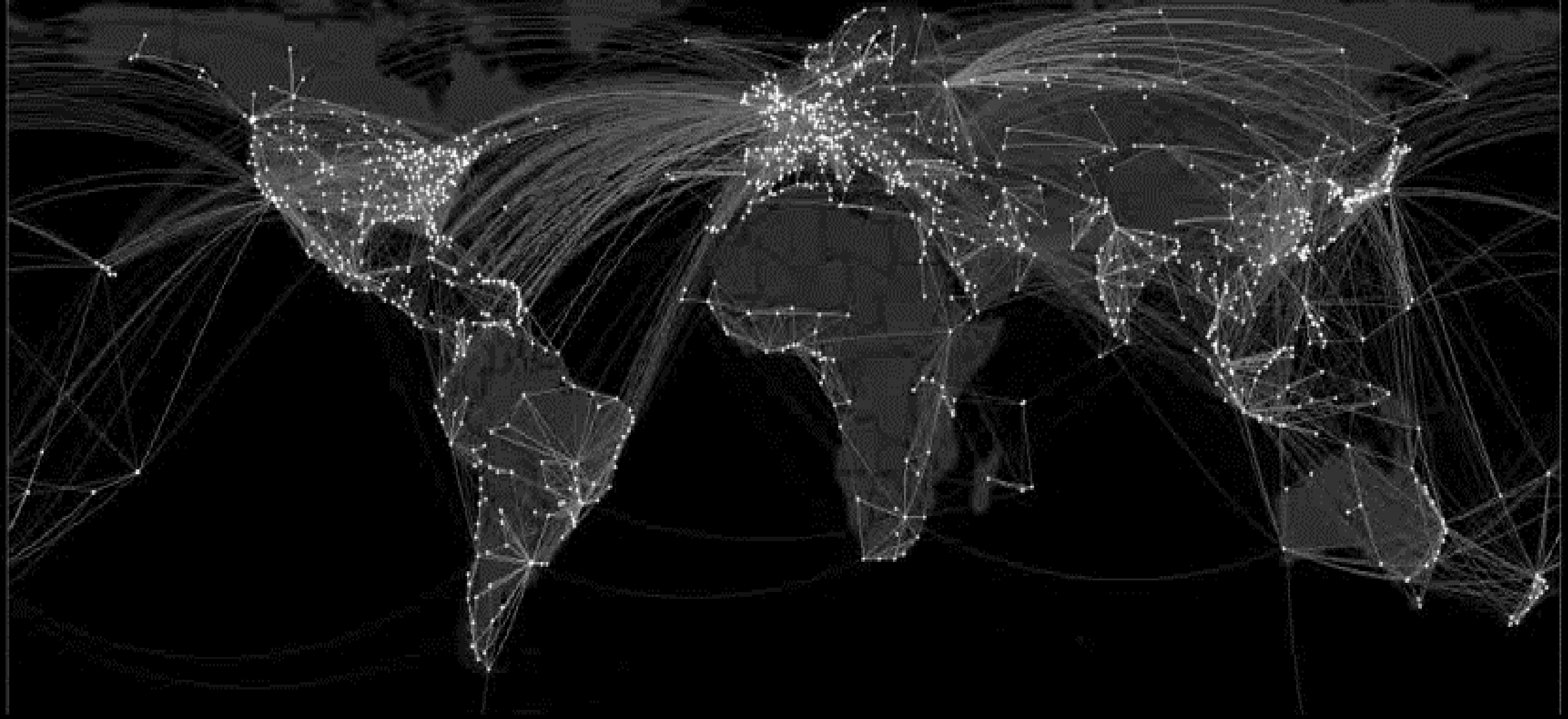
Agglomeration Economies

Nightime lights captured by satellites.

We live in a small world



Diffusion of viruses and information is fast



Brockman D., Helbing D (2013) Science

The global society is segregated



Why do we observe growing inequalities if diffusion of ideas and knowledge in social networks is very quick?

Prosperous areas next to a slum in Mumbai.

Network fragmentation and inequalities

Small-world social networks are fragmented by groups.

One of the strongest forces of group formation is prosperity.

Diffusion of innovation is unequal across network fragments.



Currarini, Pin, Jackson (2009) Econometrica

The second block

6	Oct 21	***WRITTEN EXAM 1*** ***Balázs Lengyel*** Learning and innovation in networks: knowledge sourcing and dynamics of networks	II. Innovation and Economic Geogra	[7,8]	*NO*
7	Nov 4	***Balázs Lengyel and Sándor Juhász*** Spatial and network inequalities: social capital, segregation and mixing		[9, 10]	Quiz5 HW3
8	*Nov 11*	***Balázs Lengyel*** Spatial diffusion of innovation: agents of change, interregional networks, and spatial life cycles		[11, **12**]	Quiz6
9	Nov 18	***Presentation of students' research proposal***			*NO*

[7] Fleming, L., Charles King III., & Juda, A. I (2007). Small worlds of regional innovation. *Organization Science*, 18(6), 938-954.

[8] Abbasiharofteh, M., Kogler, D. F., & Lengyel, B. (2023). Atypical combinations of technologies in regional co-inventor networks. *Research Policy*, *52*(10), 104886.

[9] Tóth, G., Wachs, J., Di Clemente, R., Jakobi, Á., Ságvári, B., Kertész, J., & Lengyel, B. (2021). Inequality is rising where social network segregation interacts with urban topology. *Nature Communications*, *12*(1), 1143.

[10] Aiello, L. M., Vybornova, A., Juhász, S., Szell, M., & Bokányi, E. (2025). Urban highways are barriers to social ties. *PNAS*, *122*(10), e2408937122.

[11] Elekes, Z., Boschma, R., & Lengyel, B. (2019). Foreign-owned firms as agents of structural change in regions. *Regional Studies*, *53*(11), 1603-1613.

[12] Lengyel, B., Bokányi, E., Di Clemente, R., Kertész, J., & González, M. C. (2020). The role of geography in the complex diffusion of innovations. *Scientific Reports*, *10*(1), 15065.

Today

1. Small-world networks

2. Benefits in small-world networks

3. Innovation in small-world networks

1. Small-world networks



Small worlds

Chains

SZÉPIRODALMI KÖNYVKIADÓ

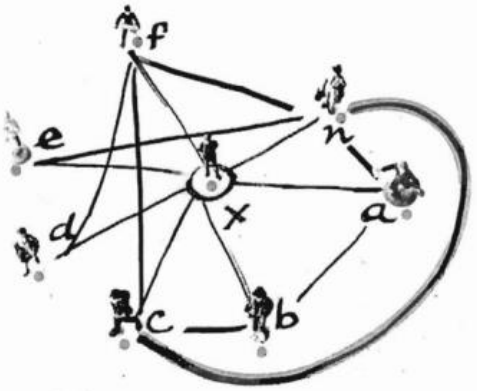
1929

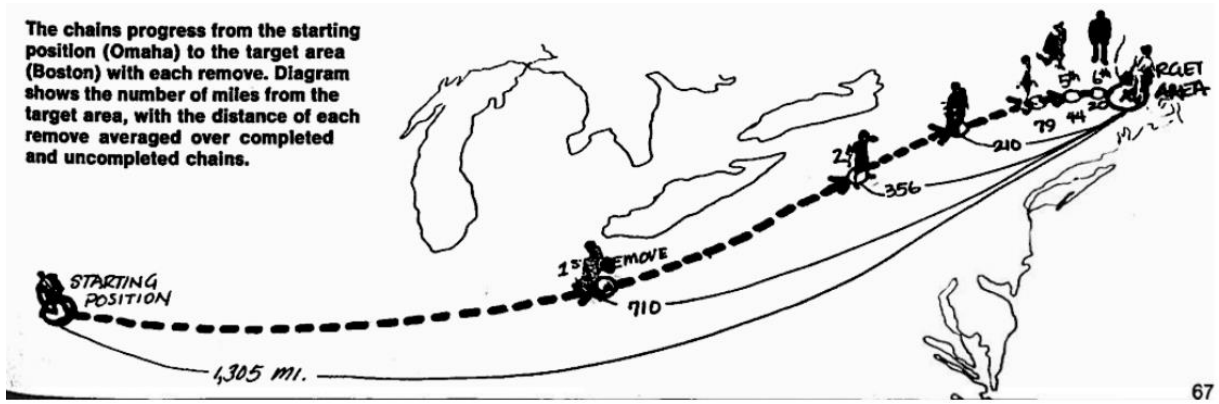
A fascinating game grew out of this discussion. One of us suggested performing the following experiment to prove that the population of the Earth is closer together now than they have ever been before. We should select any person from the 1.5 billion inhabitants of the Earth—anyone, anywhere at all. He bet us that, using no more than five individuals, one of whom is a personal acquaintance, he could contact the selected individual using nothing except the network of personal acquaintances. For example, "Look, you know Mr. X.Y., please ask him to contact his friend Mr. Q.Z., whom he knows, and so forth."

"An interesting idea!" someone said "Let's give it a try. How would you contact Selma Lagerlöf?"



Small-world networks



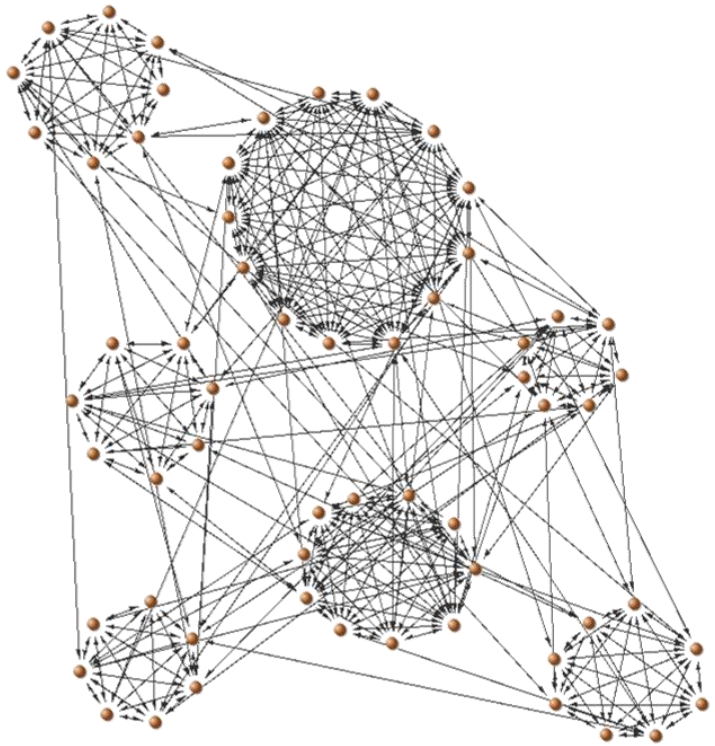


Milgram, S. (1967) The Small-world Problem. *Psychology Today*, 1, 1, 61-67.

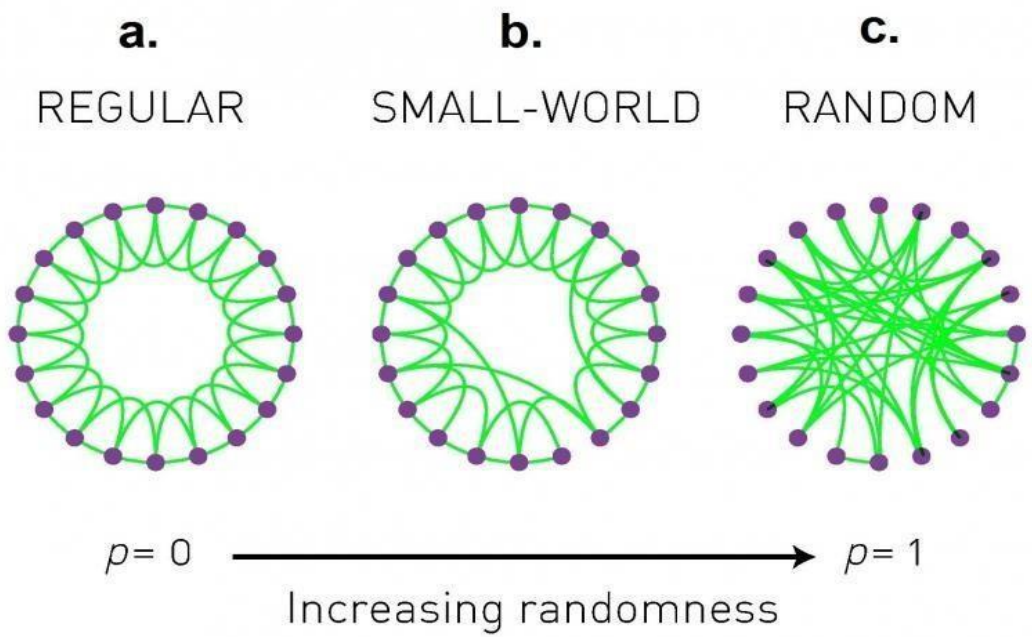


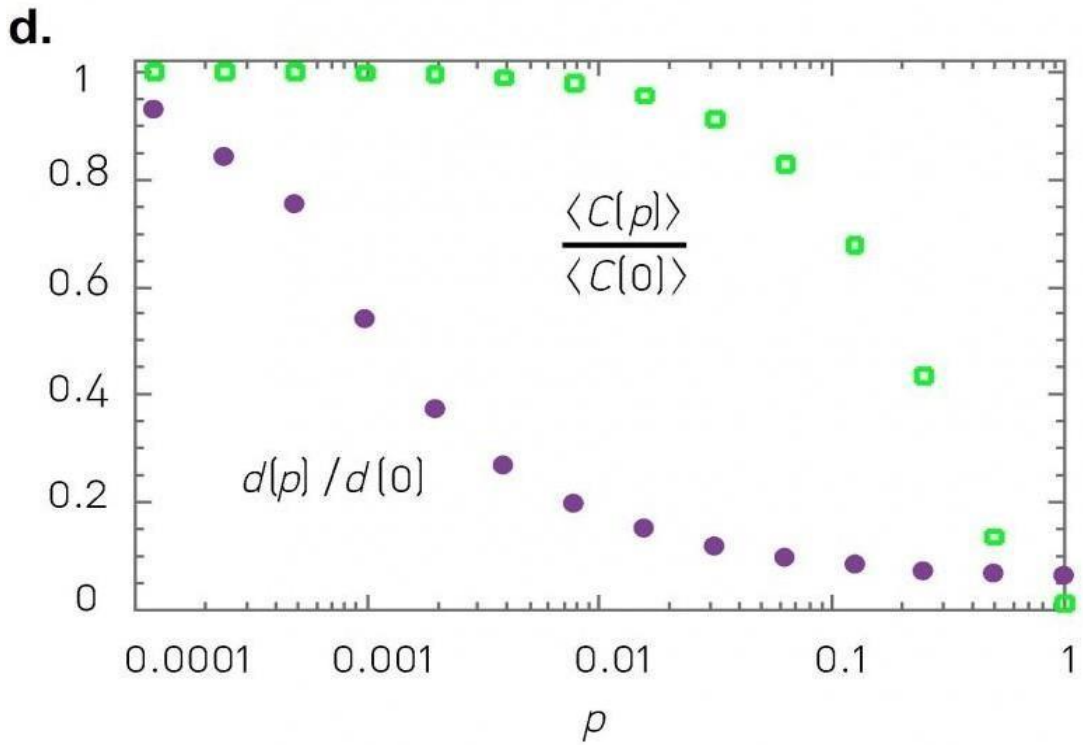
Small-world networks

1. Dense local relations / people share many friends
2. Few bridges make paths short



Watts, D.J., Strogatz, S.H. (1998) Collective Dynamics of Small-world Networks. *Nature*, 393, 440-442.


a.
b.
c.
REGULAR
SMALL-WORLD
RANDOM
p= 0
p= 1
Increasing randomness


d.
1
0.8
0.6
0.4
0.2
0
⟨C(p)⟩ / ⟨C(0)⟩
d(p) / d(0)
0.0001
0.001
0.01
0.1
1
p

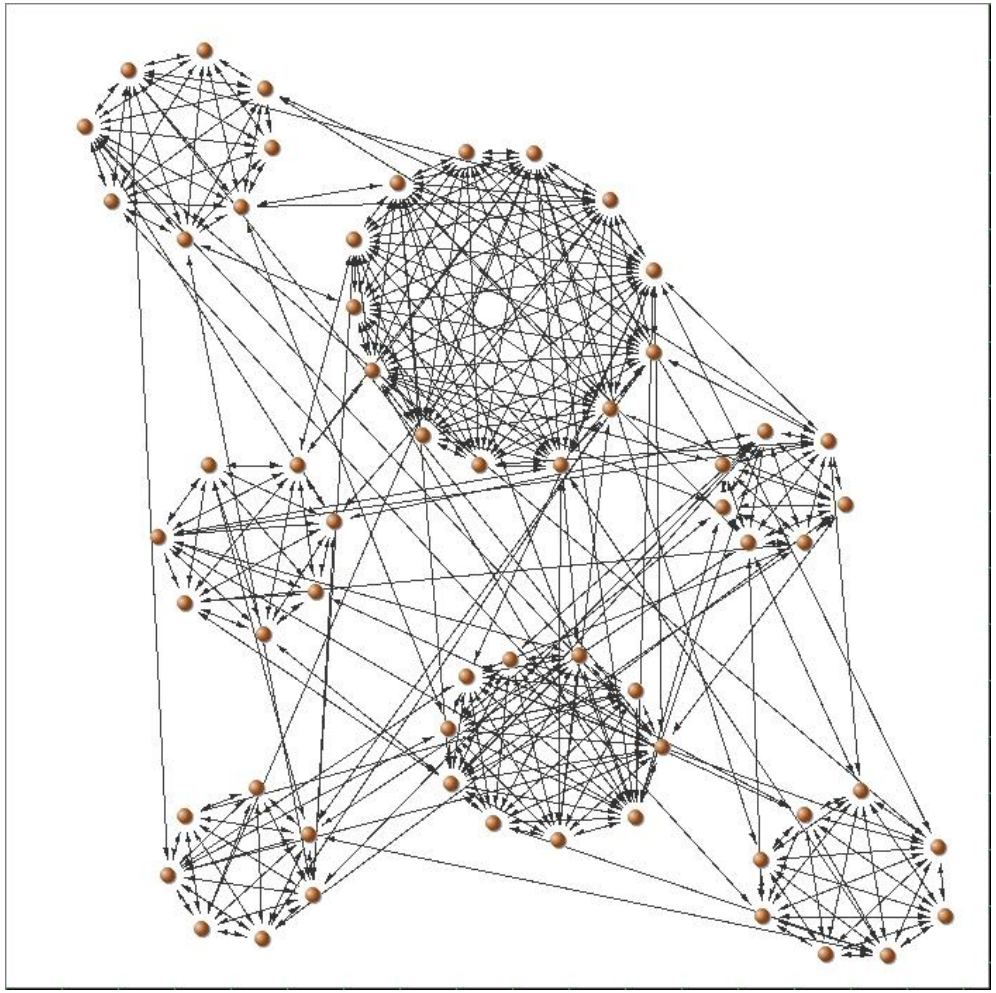
Small worlds

- Social networks cannot be described by random ties:

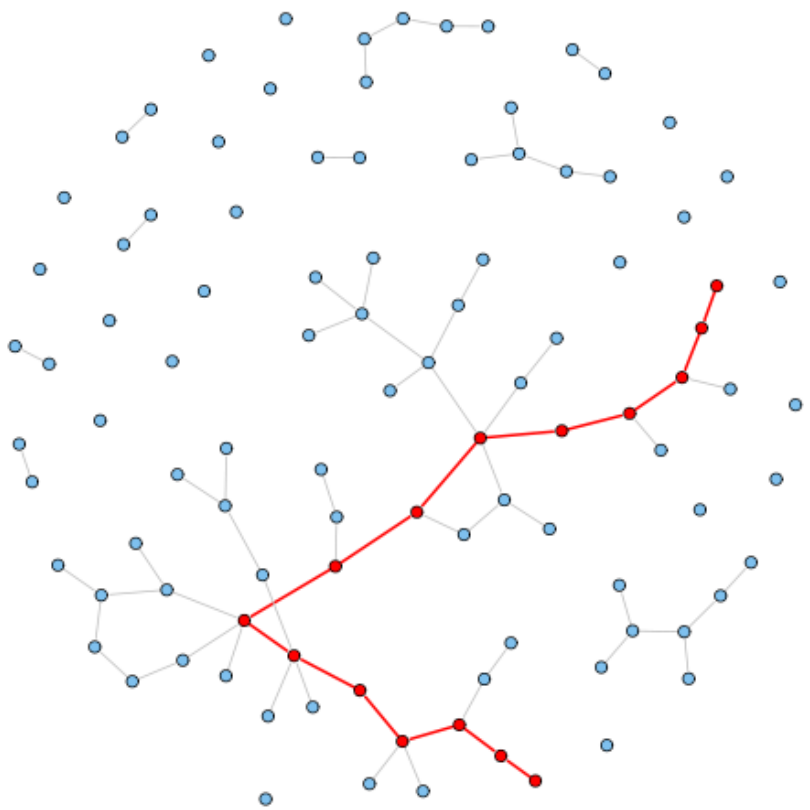
  Dense local relations and

  few bridges between isolated groups

- Small worlds
  - High clustering – due to dense local networks
  - Short paths – due to bridges

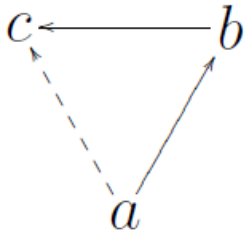


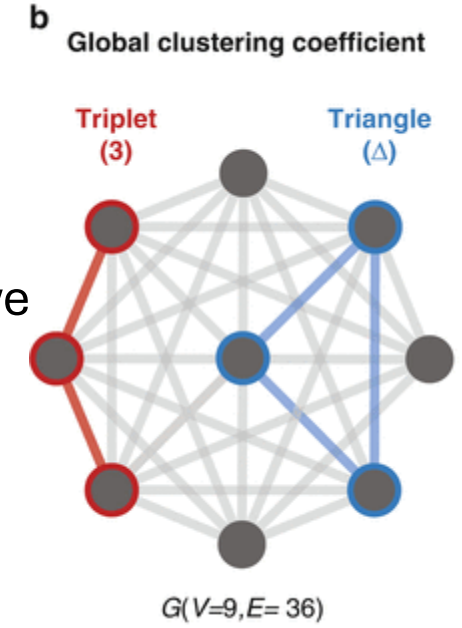
Path



- Path: The set of edges that we can get from one node to any other node
- Distance: the number of steps needed to get from one node to another in the network
- Shortest path: the path between two nodes that includes the fewest steps

Clustering





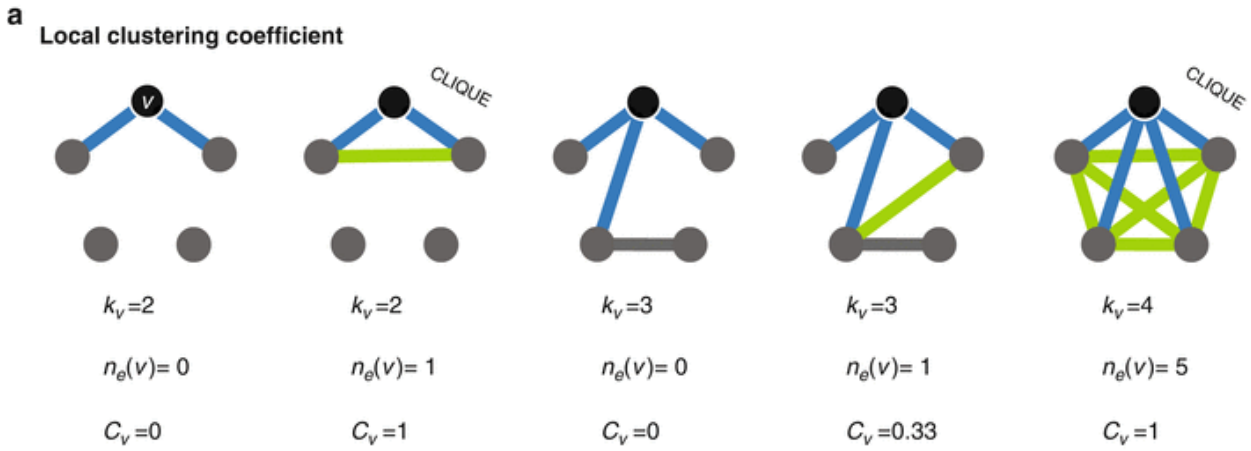
- Transitivity: if ‘a’ knows ‘b’ and ‘b’ knows ‘c’-t, then ‘a’ is likely to know ‘c’ as we
  - Note that we concentrate on tiadic relations in this measure!

- Global clustering: how many triads are closed out of all possible triads?

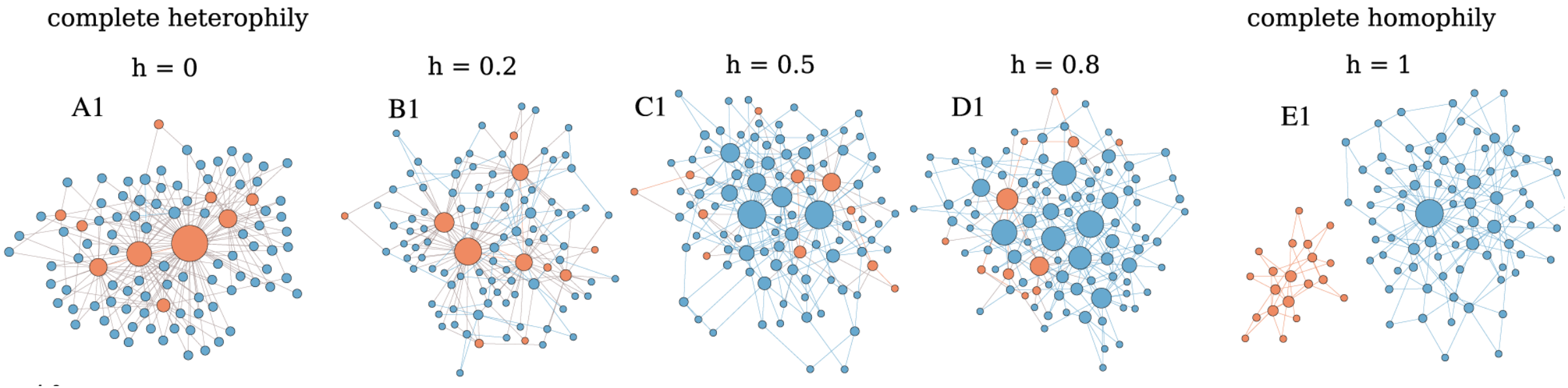
- Local clustering: for all *j,k* neighbor-pairs of *I, how many times does the j-k* edge materialize?

  (one can sum this up

  for the full network)



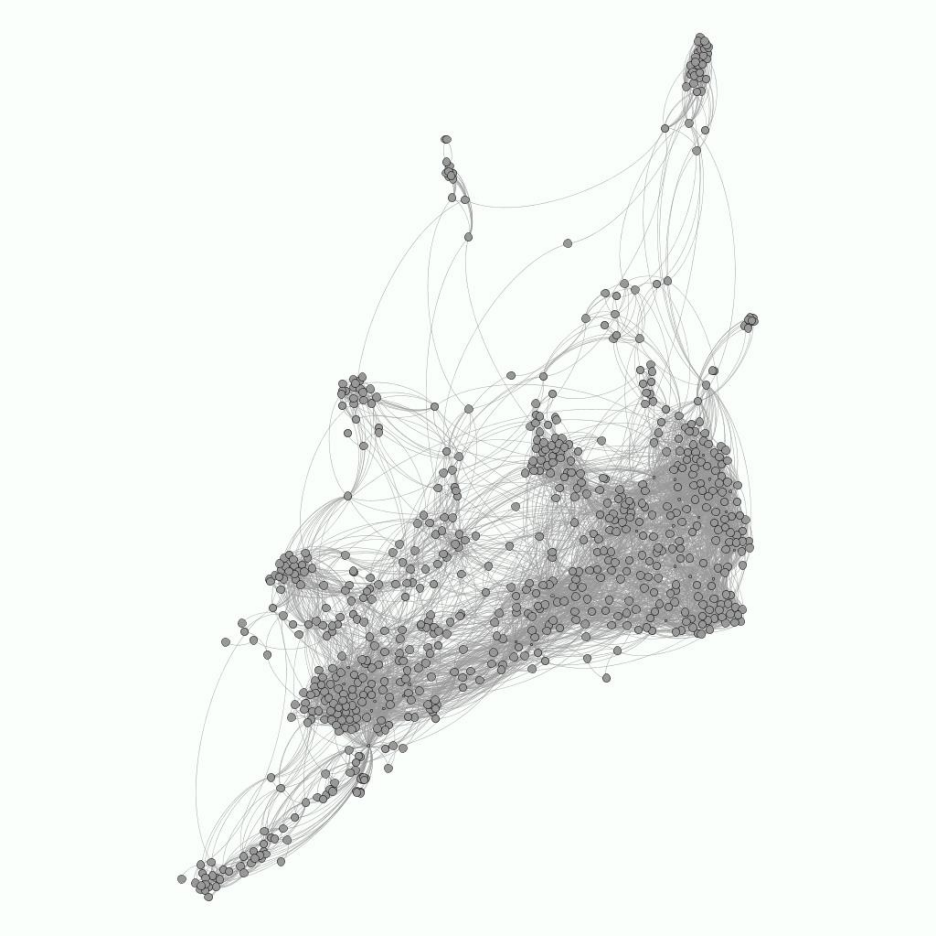
Homophily / assortativity



Modularity

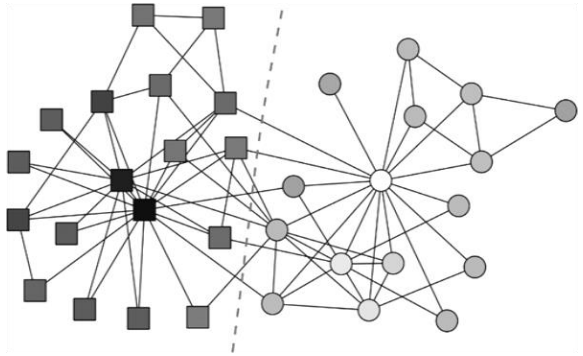
Moduls

Groups/subnetworks/moduls/communities emerge due to transitivity



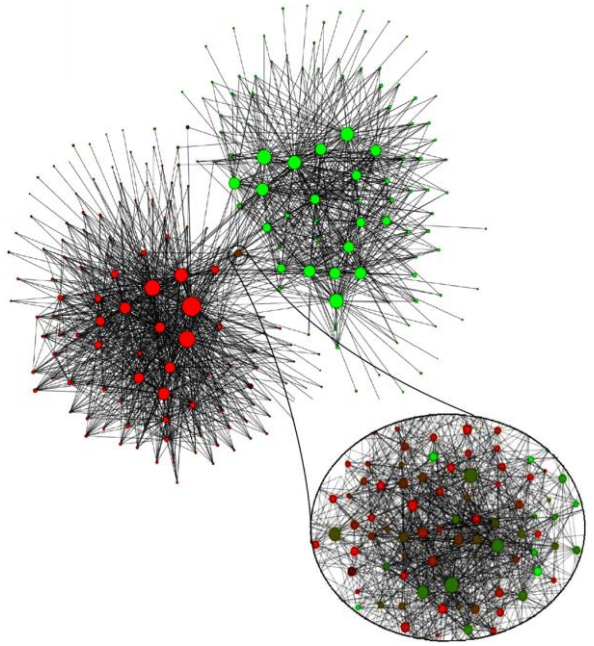
Density Hypothesis of Community detection:

Communities correspond to locally dense neighborhoods of a network



→ Karate Club: Breakup of the club

→ Belgian Phone Data: Language spoken



2. Benefit in small-world networks

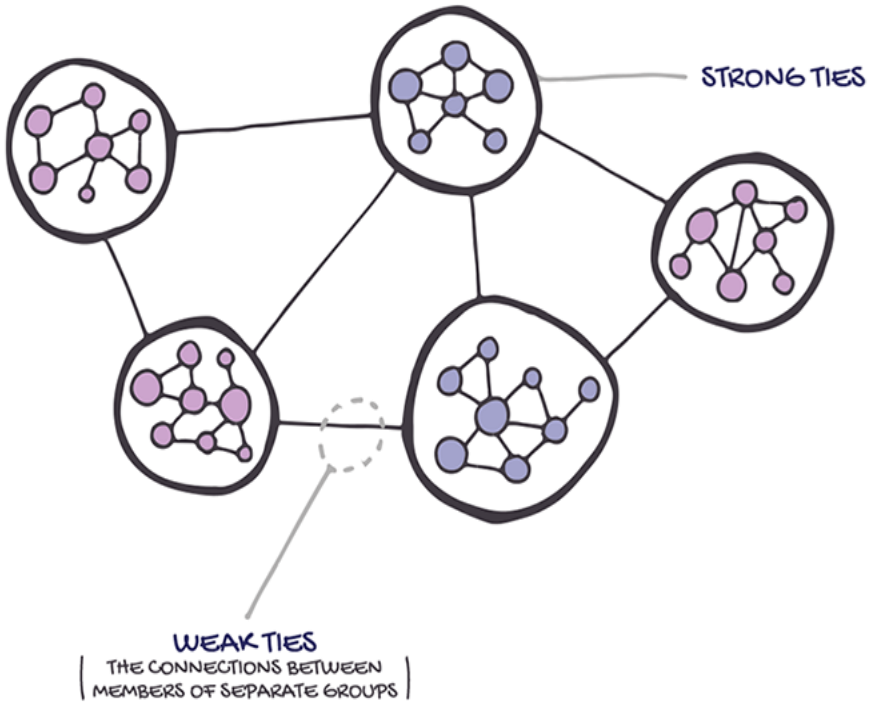
Weak ties and Brokers

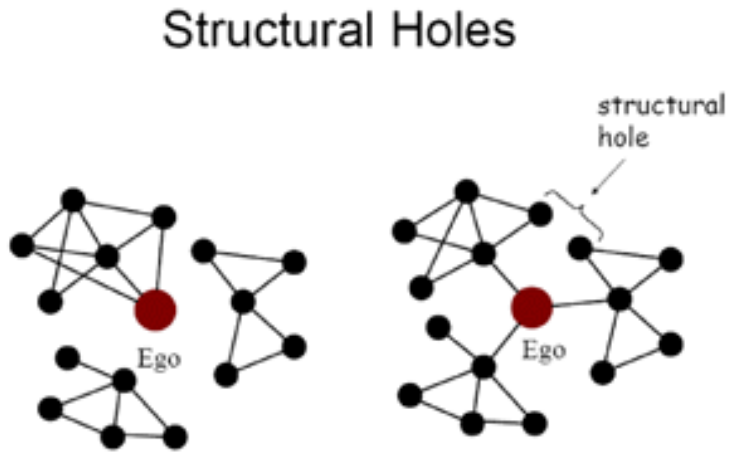
Weak ties (Granovetter, 1973):

- Valuable information comes from those contacts who are in occasional/not frequent relation with us.

Advantage of Brokers (Burt, 1992)

- Information is homogenous and redundant in dense clusters.
- In case information is not overlapping between two loosely knit clusters, there is a structural hole between these clusters.
- Brokers:
  - tertius gaudens -> control of flows
  - tertius iungens -> establish links

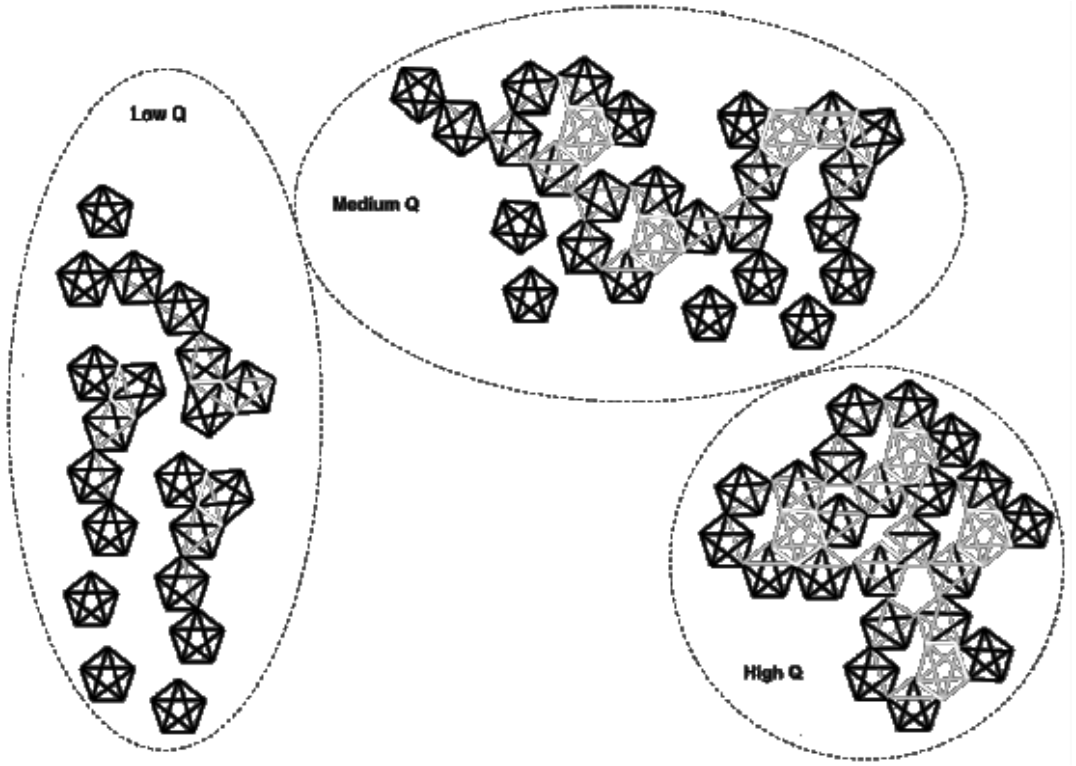




Optimal small worldliness: combination of diverse and cohesive networks

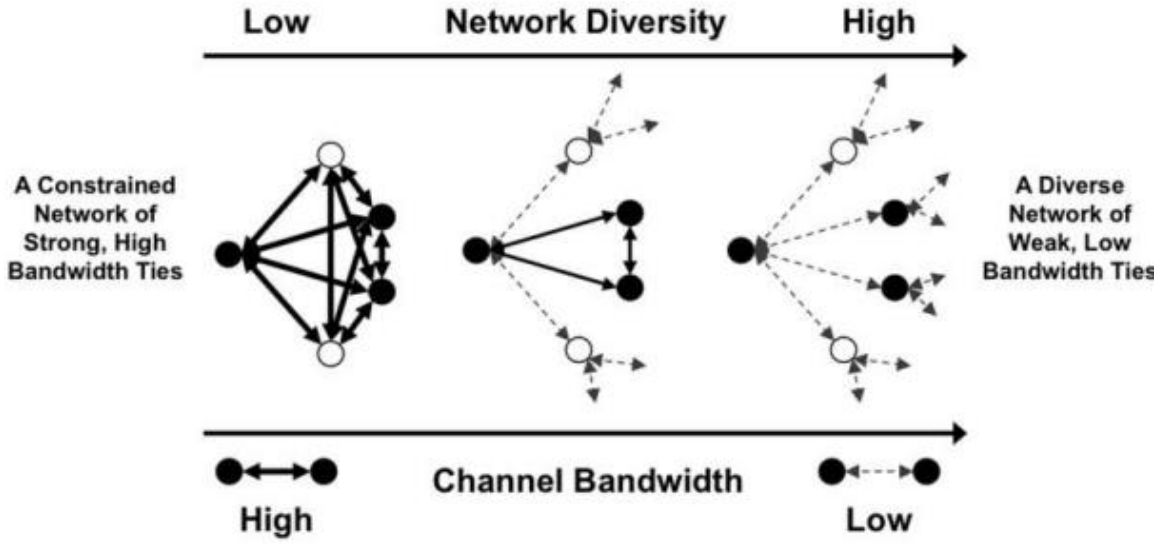
Uzzi and Spiro (2005) American Journal of Sociology

- Small-world Q: high transitivity and short paths
- Broadway musical example: high quality performance needs medium small-worldliness



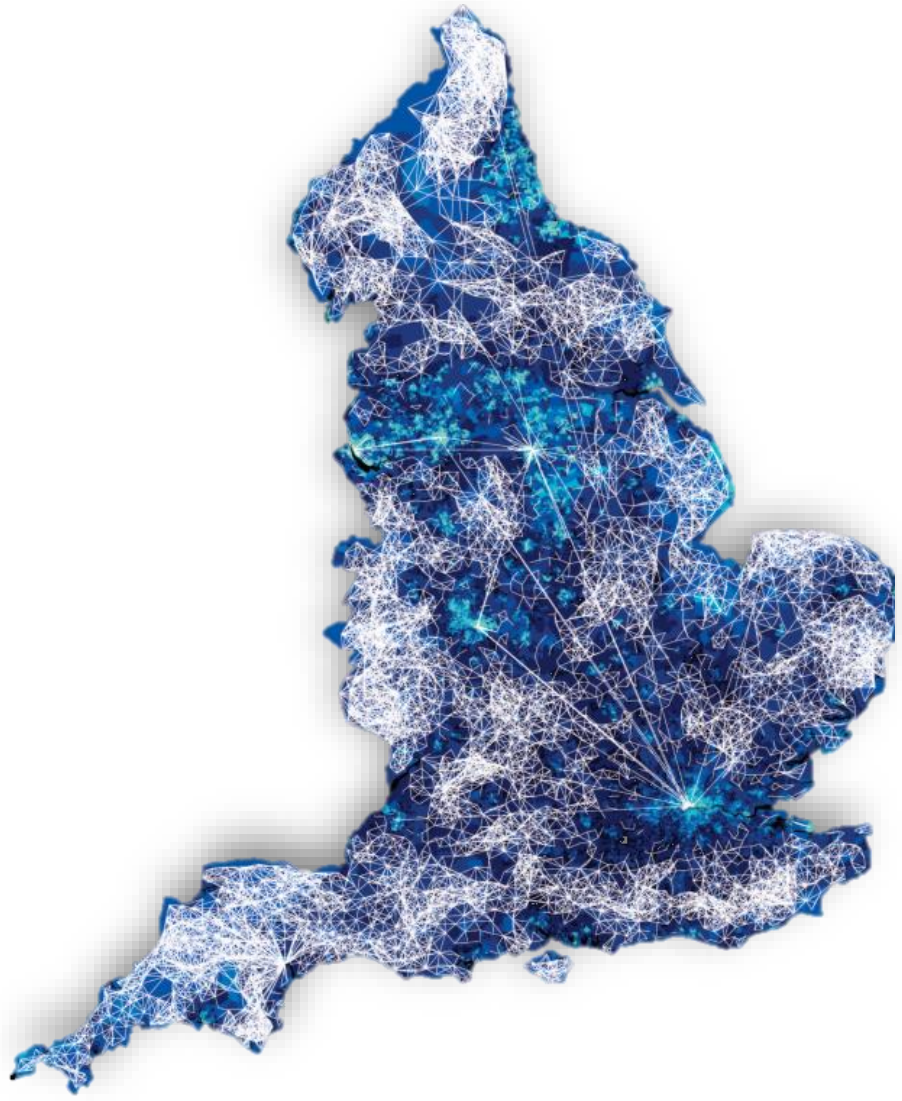
Benefits in small-world networks

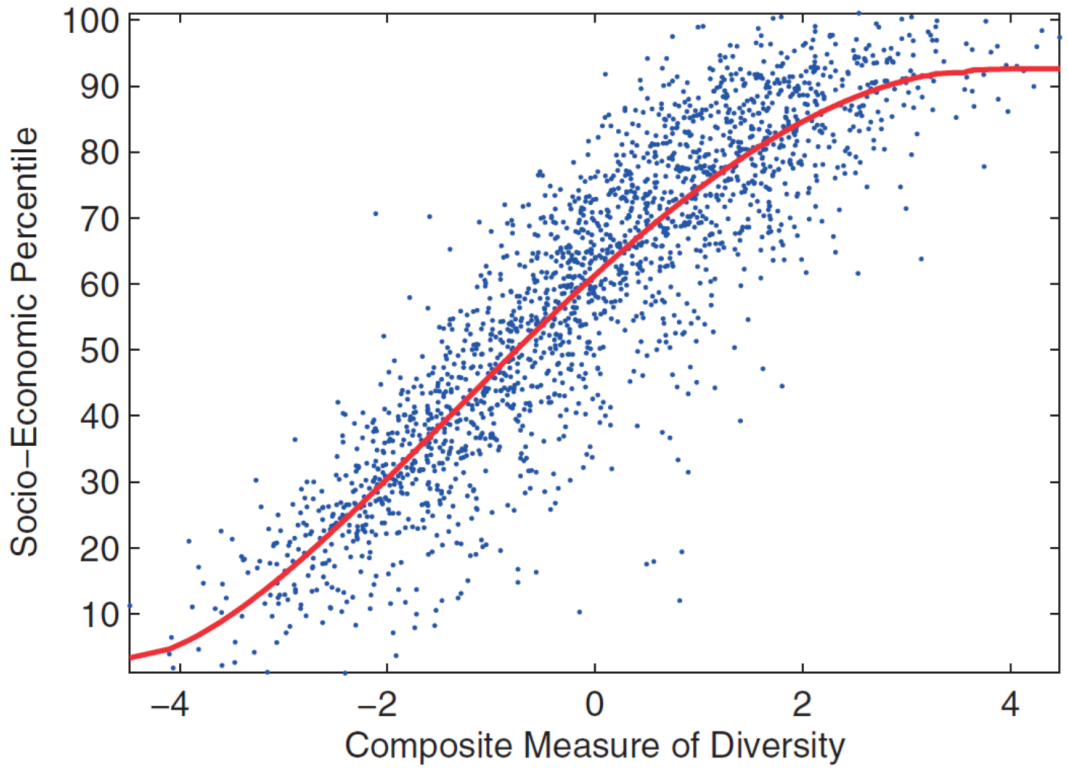
- Cliques / strong ties help us to develop trust, and process information (Coleman, 1988)
- Bridges / weak ties provide us with diverse information (Granovetter, 1973; Burt, 1992)
- A combination of them are needed for success (Uzzi and Spiro, 2005)



Aral, S. (2016) The Future of Weak Ties. *American Journal of Sociology*, 121, 6, 1931-1939.

Social networks, geography and wealth





Eagle-Macy-Claxton (2010) Science

Network dynamics in clusters (Juhász and Lengyel JoEG 2017)

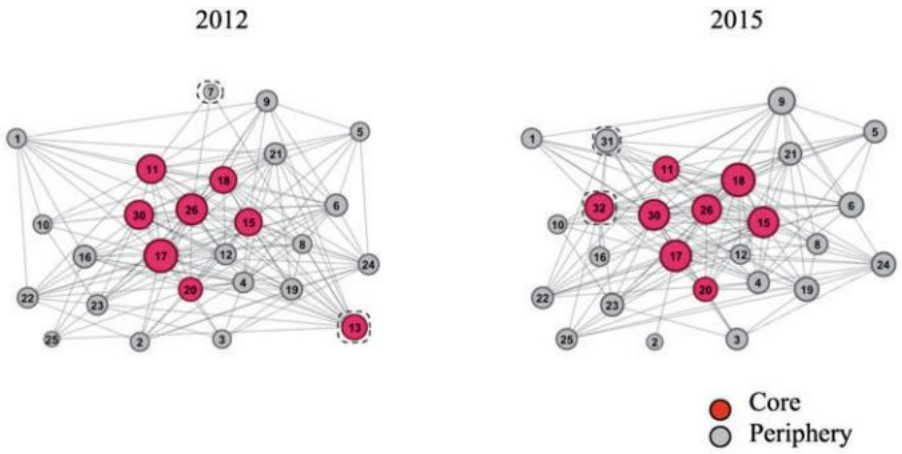
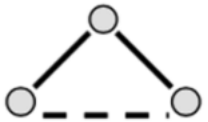
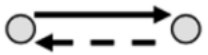
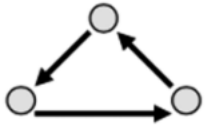
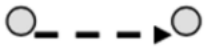


Table 2. Tie changes considered by the evaluation, creation and endowment functions

	Evaluation		Number of ties	Creation		Number of ties	Endowment		Number of ties
	t	$t+1$		t	$t+1$		t	$t+1$	
Creation	$i \quad j$	$i \rightarrow j$	71	$i \quad j$	$i \rightarrow j$	71			
Persistence	$i \rightarrow j$	$i \rightarrow j$	110				$i \rightarrow j$	$i \rightarrow j$	110
Termination	$i \rightarrow j$	$i \quad j$	113				$i \rightarrow j$	$i \quad j$	113
No ties	$i \quad j$	$i \quad j$	462	$i \quad j$	$i \quad j$	462			
Odds ratio			181/575			71/462			110/113

Source: Author's own construction based on Ripley et al. (2017).

Table 3. Operationalization of structural, dyadic and firm-level variables

Structural variables			
	Description	Formula	Visualization
Triadic closure	Tendency toward triadic closure when two knowledge ties existed in the previous period	$T_i = \sum_{j,h} x_{ij} x_{ih} x_{jh}$	
Reciprocity	Tendency of mutual knowledge exchange	$R_i = \sum_j x_{ij} x_{ji}$	
Cyclicity	Tendency of knowledge exchange in cycles	$C_i = \sum_{j,h} x_{ij} x_{jh} x_{hi}$	
Density	Overall tendency of actors to ask advices	$D_i = \sum_j x_{ij}$	
Dyadic variables			
Geographical proximity	Physical distance of firms subtracted from the maximum distance in the sample		
Cognitive proximity	Number of digits two firms share in common in their NACE 4 codes		
Triadic closure X Cognitive proximity	Number of common third partners multiplied by the number of digits two firms share in common in their NACE 4 codes		
Firm-level variables			
External knowledge ties	Number of knowledge linkages outside the region		
Age	Number of years since establishment		
Ownership	Equals *1* if foreign and *0* if domestic		
Employment	Total number of employees		

3. Innovation in small-world networks

Research Policy 52 (2023) 104886



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Atypical combinations of technologies in regional co-inventor networks

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ISSN 1047-7039 | EISSN 1526-5455 | 07 | 1806 | 0938

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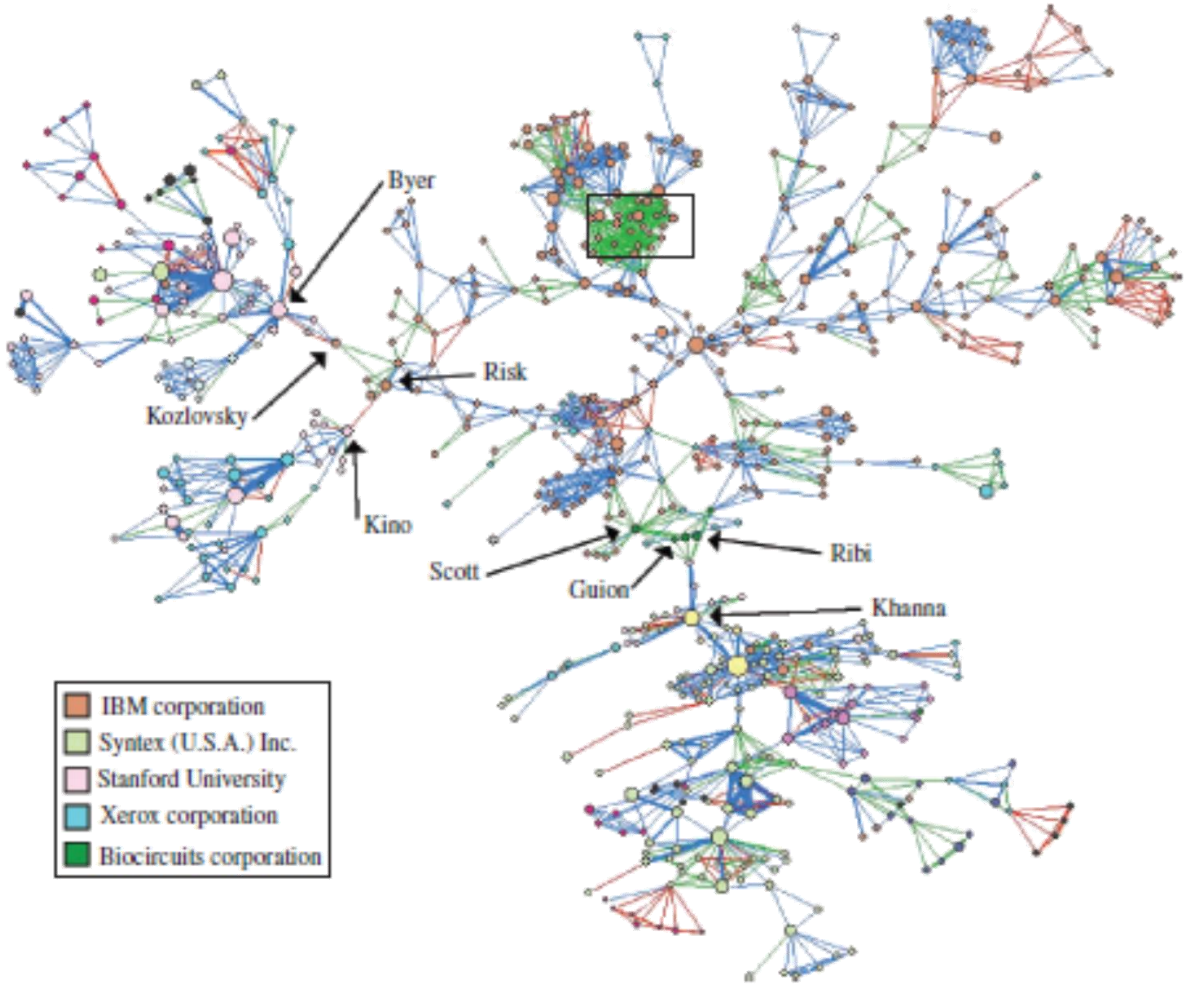
Small Worlds and Regional Innovation

Lee Fleming
Harvard Business School, Morgan Hall 485, Boston, Massachusetts 02163, lfleming@hbs.edu

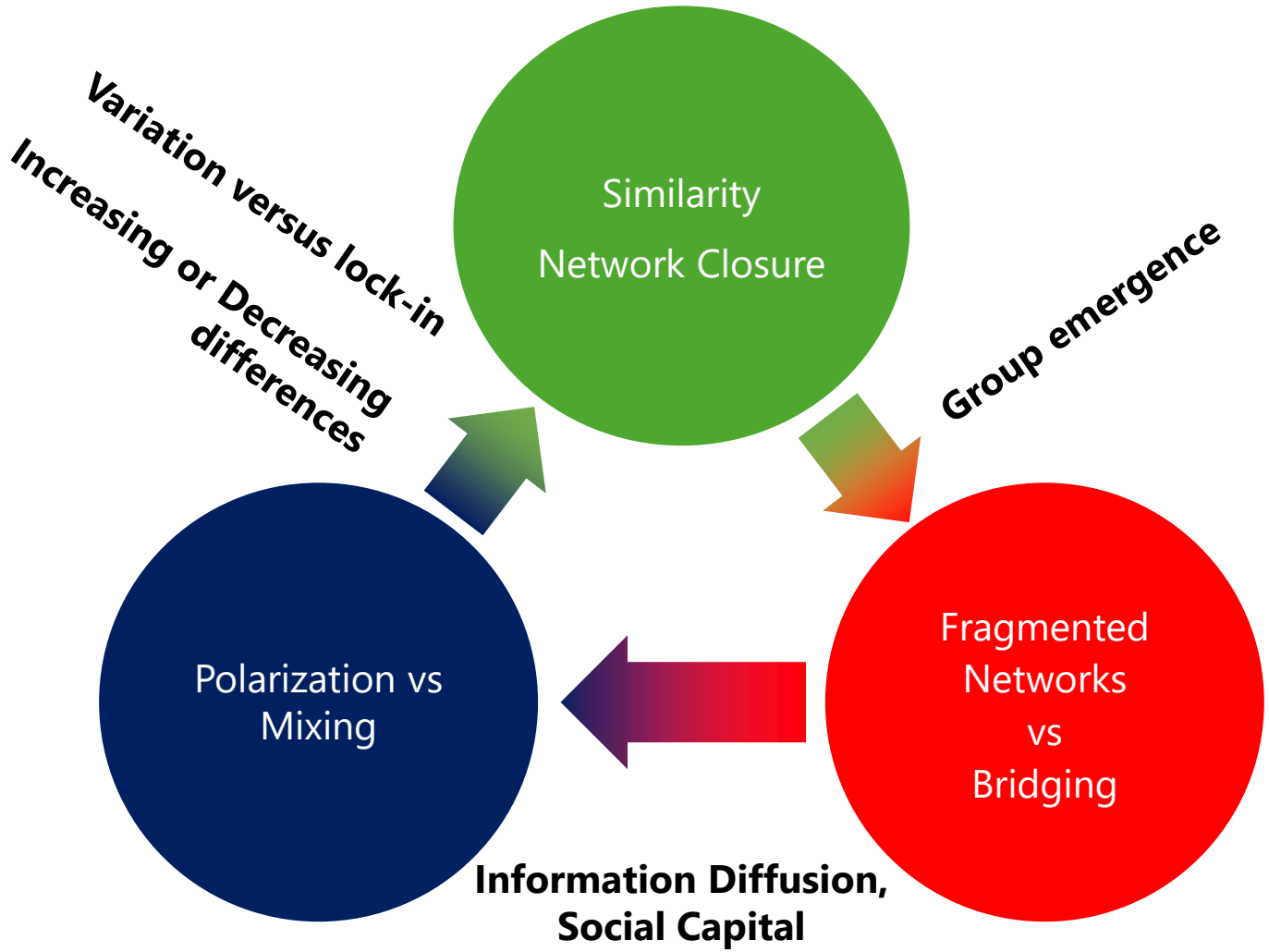
Charles King III
Greylock McKinnon Associates, One Memorial Drive, Suite 1410, Cambridge, Massachusetts 02142, and
Pleiades Consulting Group, Inc., P.O. Box 711, Lincoln, Massachusetts 01773, cking@gma-us.com

Adam I. Juda
Google, Inc., 76 Ninth Avenue, 4th Floor, New York, New York 10011, juda@goggle.com

Short access in the co-inventor network speeds up information flow and induces the combination of knowlegde.



A Network Dynamics Framework



The role of knowledge: diversity vs specialization

Radical novelty needs diverse knowledge (Florida et al. 2017, Glaeser et al. 1992).

Specialization can also facilitate innovation (Beaudry and Schiffauerova, 2009; Lobo and Strumsky, 2008) when critical masses of experts specialized in distinct pieces of knowledge (Castaldi et al., 2015) are connected through knowledge transfer mechanisms (Berkes and Gaetani, 2020).

Boston biotechnology example (Cooke 2002, Powell et al. 1996): distinct local critical masses in engineering and biology connected by social ties.

The role of knowledge: network dynamics

Similar specialization in knowledge increases link probability.

- Local collaboration networks can become too cohesive and locked-in into technologies – Grabher (1993), Boschma & Frenken (2010), Giuliani (2013), Balland et al. (2016)

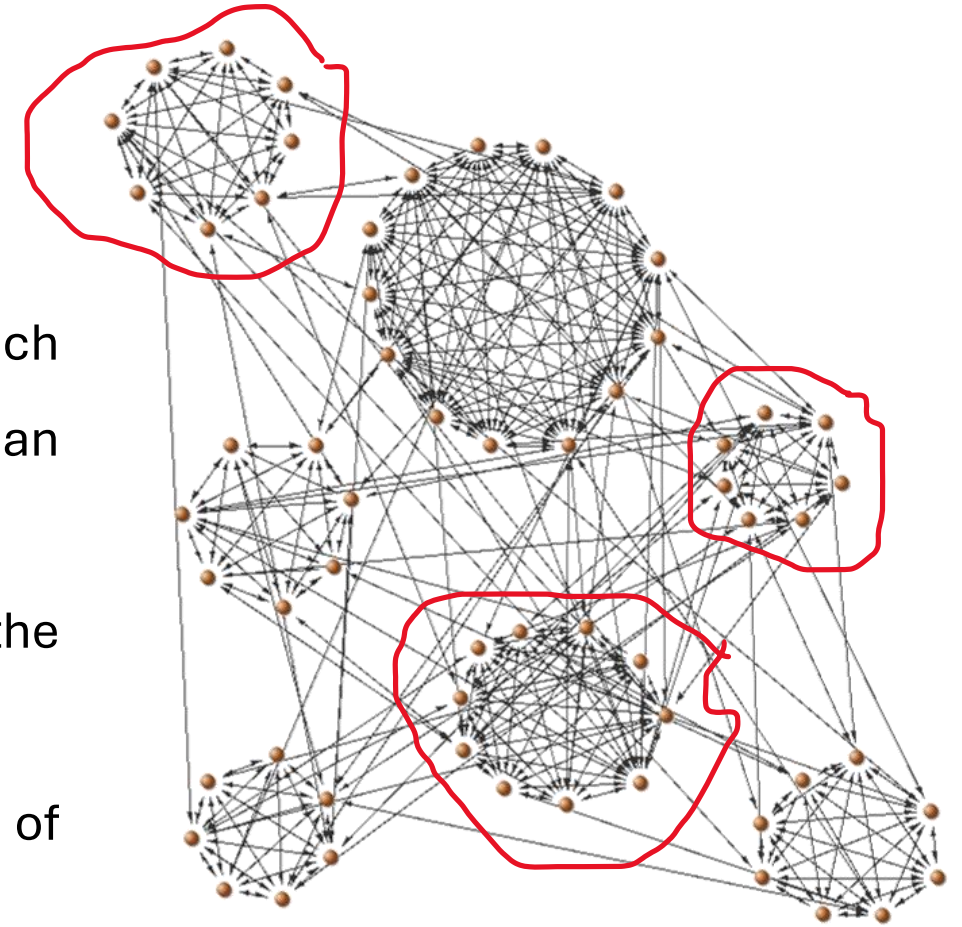
Inter-regional links can introduce novelty

- Inter-regional links combined with dense local networks
  – theory: Bathelt et al. (2004), Glückler (2007);
  – empirical evidence: Breschi and Lenzi (2016), Eriksson and Lengyel (2019); Kogler et al. (2023)
- Local variation – Juhász and Lengyel (2018)

Problem: the impact of knowledge and network dynamics are difficult to disentangle.

Solution: network communities

Network communities: cohesive segments of networks that are loosely connected to other communities.



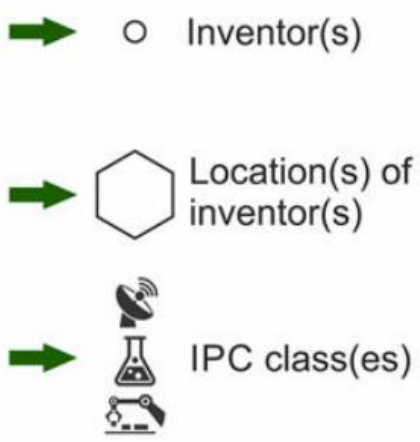
Advantages:

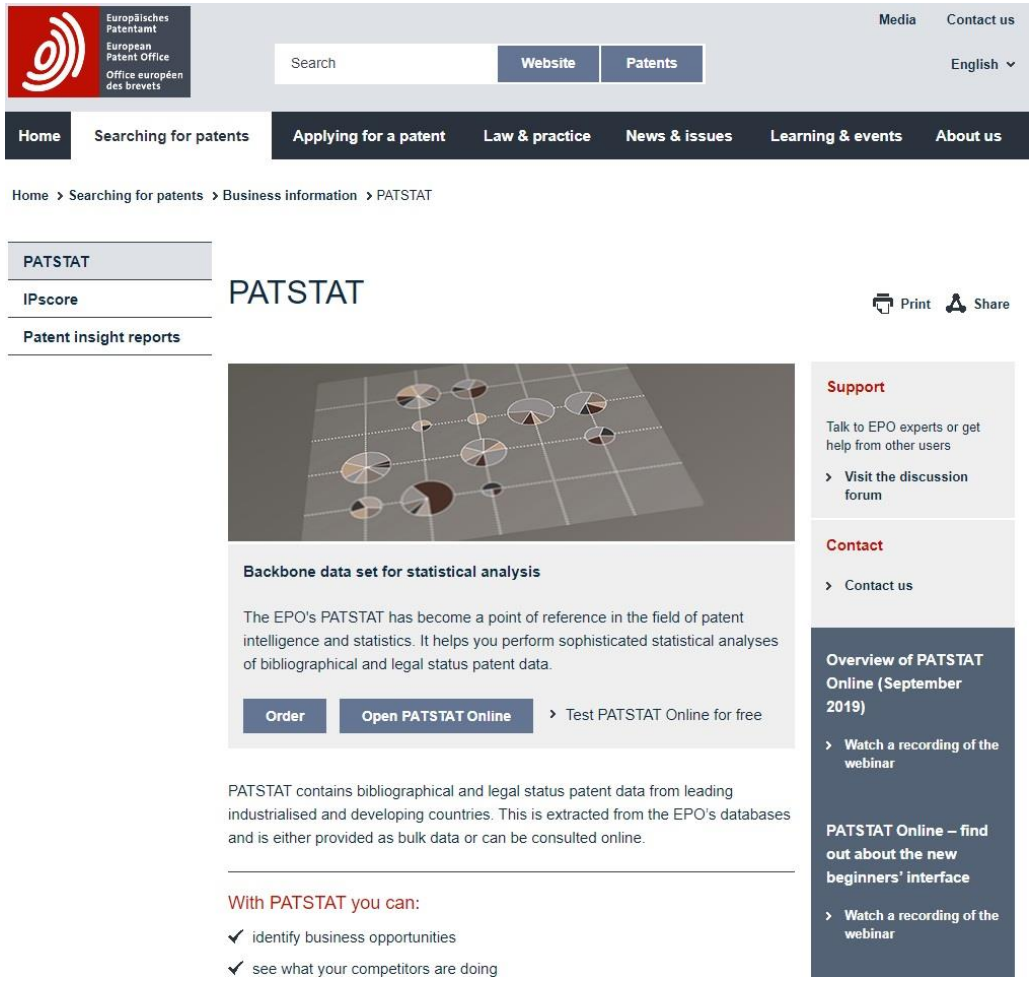
- Small-world networks can be decomposed to communities such that bridges are inter-community ties (Girwan and Newman 2002)
- Community detection only considers the structure of the network (Fortunato 2010) – Triadic Closure effect
- One can measure the technological specialization of communities – Technological Proximity effect

Data from European Patents

- The European Patent Office (EPO) PATSTAT database
- Creating co-inventor network in seven non-overlapping 5-year time-windows for each NUTS2 region







Tóth et al. (2021) European Planning Studies

Atypical patents as radical innovation

SHARE

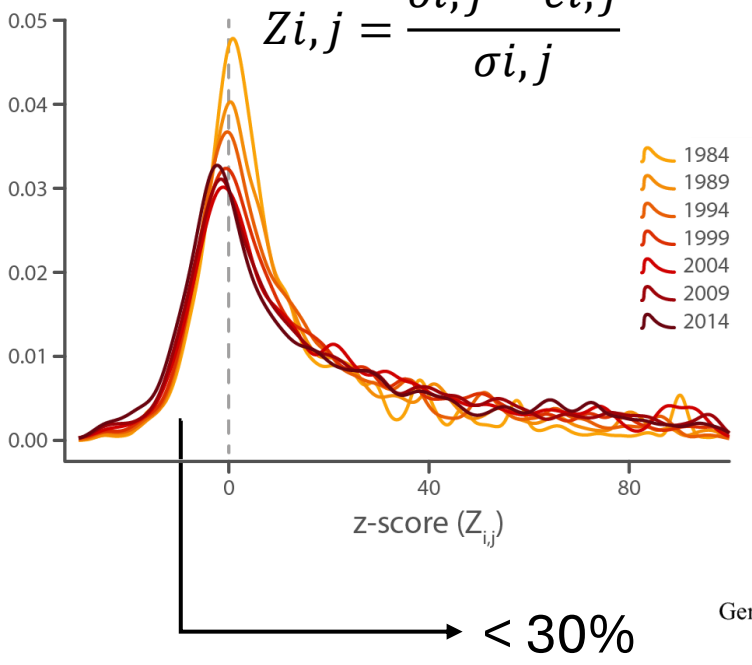
REPORT

Atypical Combinations and Scientific Impact

Brian Uzzi[1,2], Satyam Mukherjee[1,2], Michael Stringer[2,3], Ben Jones[1,4,*]
+ See all authors and affiliations

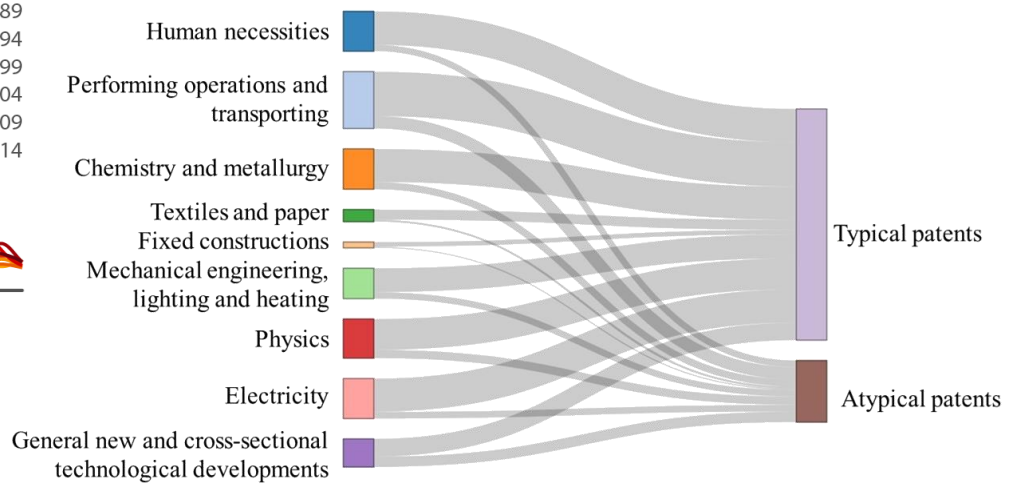
Science 25 Oct 2013:
Vol. 342, Issue 6157, pp. 468-472
DOI: 10.1126/science.1240474

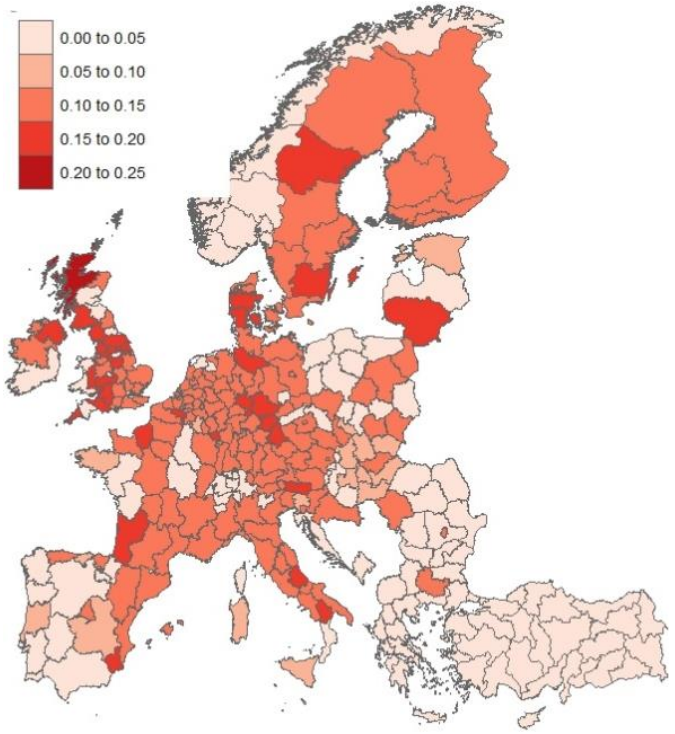
$$Zi,j = \frac{oi,j - ei,j}{\sigma i,j}$$



Negative values represent atypical combination of classes

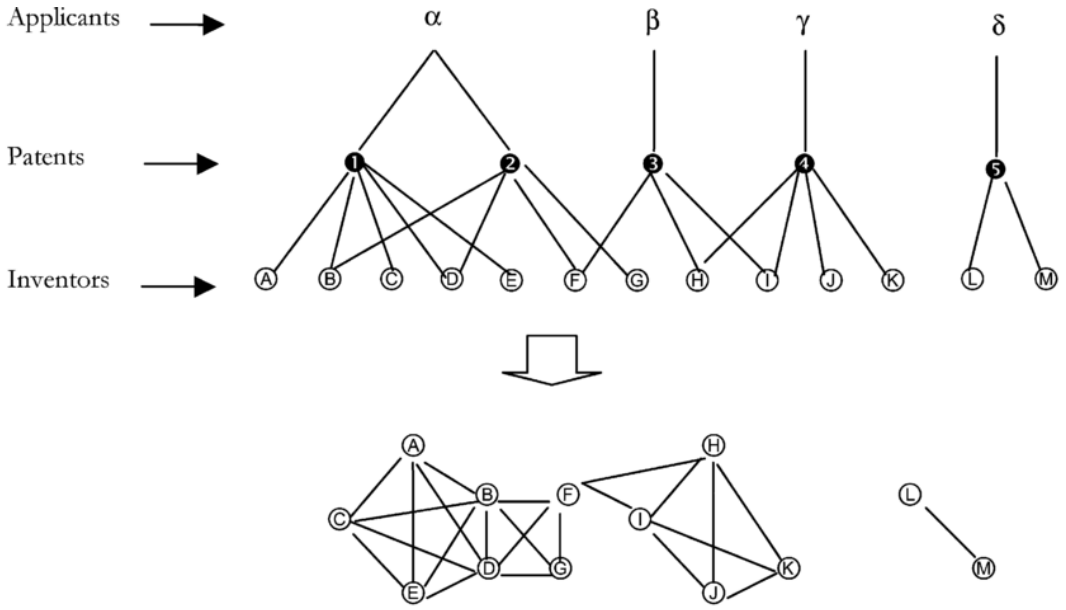
Atypical patents concentrate in cities (Mewes 2019).

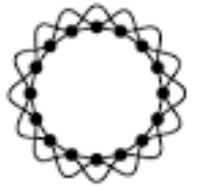
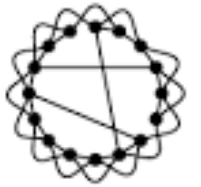




Dependent variable: share of atypical patents in EU regions, to avoid the effect of scale.

Network variables 1



			
Network	Lattice, Ordered	Small World	Random, Disordered
Clustering Coefficient	High	High	Low
Mean Path Length	Long	Short	Short

Small-worldness (Neal, 2018)

1. Average path length / compared to a random network
2. Clustering / compared to lattice

$$\omega = \frac{L_r}{L} - \frac{C}{C_l}$$

$$SMALLWORLDNESS = 1 - |\omega|$$

Small worldness ranges from 0 to 1.

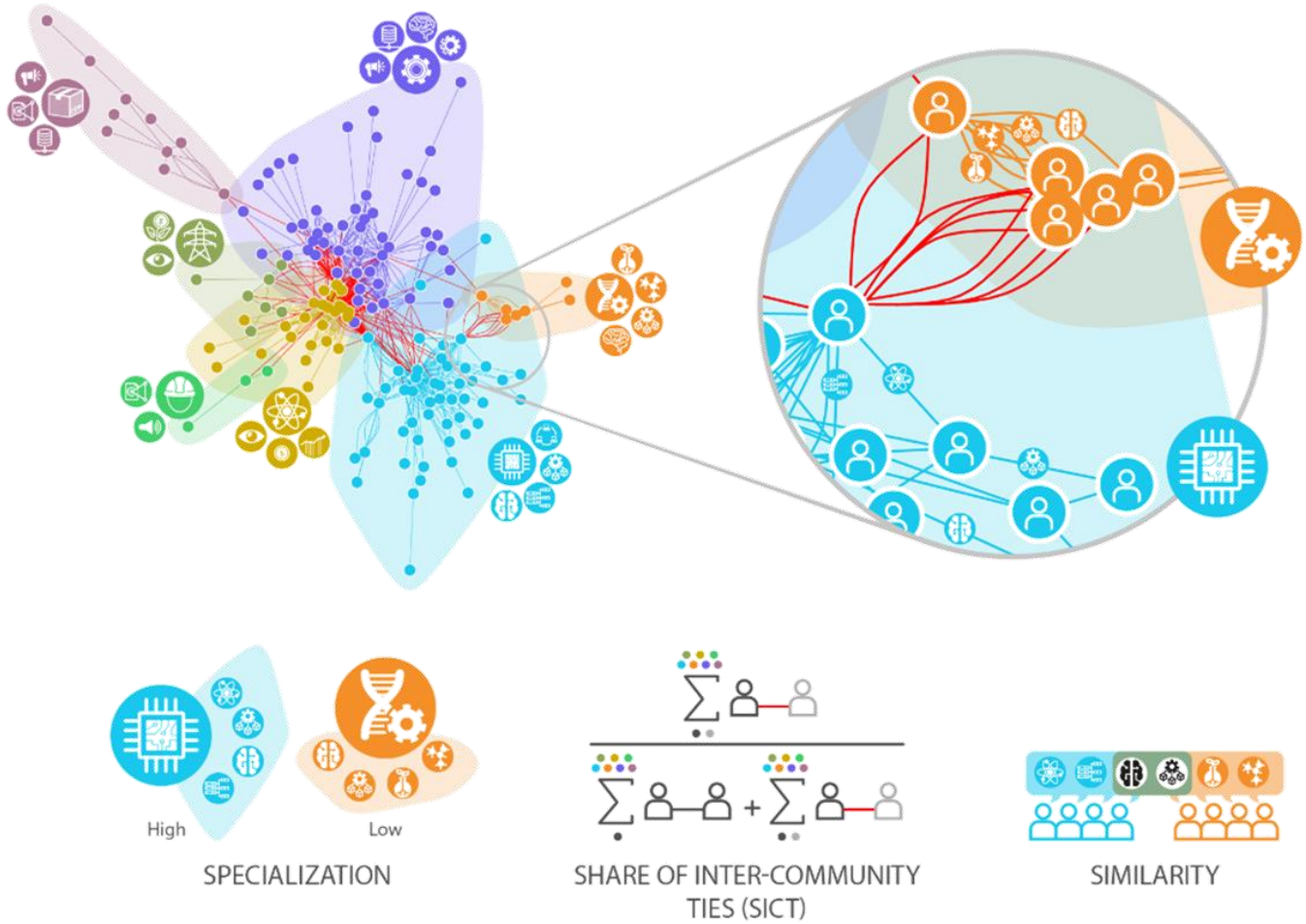
Breschi et al. (2021) Research Policy

Network variables 2

Network communities are detected by the Louvain method

Regional-level indicators

1. Specialization: median level of technology HHI in network communities (not correlated with community and inventor numbers)
2. SICT: share of inter-community ties compared to randomly rewired networks
3. Similarity: median level of technology correlations (Spearman) across connected communities



Results

$$Y_{r,t} = \alpha + \beta_1 SMALLWORLDNESS_{r,t-1} + \beta_2 SPECIALIZATION_{r,t-1} + \beta_3 SICT_{r,t-1} + \beta_4 SIMILARITY_{r,t-1} + \beta_5 N_{r,t-1} + \beta_6 Z_{r,t-1} + \varphi_t + \mu_r + \varepsilon_{r,t}$$

Controls:
1. Average Related Density (van de Wouden and Rigby 2019)
2. Complexity (Hidalgo and Hausmann 2009)
3. Density (Bergé et al., 2018; Breschi and Lenzi, 2016)
4. Isolate (Lobo and Strumsky 2008)
5. Community
6. Population
7. Interregional ties

Dependent variable: share of atypical patents	
SMALLWORLDNESS	0.0115***
	(0.0039)
SPECIALIZATION	0.0405**
	(0.0188)
SICT	0.1041**
	(0.0523)
SIMILARITY	-0.0342***
	(0.0105)
Region FE	YES
Time FE	YES
N	1,526
R^2	0.4118